



Malik Nauman Rauf

CONTACT INFORMATION

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RESEARCH INTERESTS

My research interests include flight physics, control systems and engineering, with a focus on model predictive control and adaptive control, as well as optimal control. I am also deeply interested in the modeling of uncertain and partially controllable systems, stochastic systems, astrodynamics, and astrophysics.

EDUCATION

Georgia Institute of Technology, Atlanta, Georgia US

M.S. Aerospace Engineering, August 2024 - Present

- Grade: 4.00/4.00
- Fulbright Scholar
- Primary Area: Flight Mechanics and Control
- Courses: [AE 6530](#), [MATH 6579](#), [AE 8803 VAM](#), [ECE 6553](#), [MATH 6338](#), [AE 6580](#), [ECE 6271](#), [MATH 6307](#), [MATH 7338](#), ECE 8903
- Course Projects:
 - “Implementation of Q-learning for Multi-Agent Adversarial Games”; Python; AE 8803 Optimal Control for Learning and Games
 - “Safety Promoting Nearly-Optimal Neural Lyapunov Controllers”; MATLAB; AE 6580 Nonlinear Control

Institute of Space Technology, Islamabad, Islamabad Capital Territory PK

B.S. Aerospace Engineering, September 2018 - July 2022

- Grade: 3.93/4.00 (Summa Cum Laude)
- Valedictorian, Dean’s Honor List
- Capstone Project: “Designing a Framework for Damage Tolerance Analysis of Metallic Structures
- Electives: Flight Control Systems, Spacecraft Dynamics and Controls, Guidance and Navigation of Aerospace Vehicles

HONORS AND AWARDS

Selected for Wolfram Summer Research Institute Program, 2026

Selected for the Georgia Tech Boeing FLT Program, 2025

Fulbright Scholarship for Masters, 2024

National Award of Pakistan - International Astronomy, and Astrophysics Competition, 2024

Gold Honor - International Astronomy, and Astrophysics Competition, 2024

Achieved Academy Participant, 2024

Teknofest Turkey 2022: Winner of Black Sea Competition, Honorable Mention Prize in International Free Mission vehicle Category

Six Provisional Asteroid Discoveries - International Astronomical Search Collaboration, 2023

Institute of Space Technology: Graduated Summa Magna Cum Laude, 2022; Presidential Gold

Medal for Highest grade, 2022; Vice-Chancellor's Gold Medal for best Thesis, 2022; Brigadier Dr. Atiq-ur-Rehman Graduation Prize for Best Student of the Year 2022; Valedictorian of the Year 2022; Merit Scholarship Holder, 2018-2022
 Silver Honor - International Astronomy, and Astrophysics Competition, 2021
 STEP Scholarship Holder for Bachelors, 2018

COMPUTER SKILLS

- Languages: MATLAB, Python, C++, C#, L^AT_EX, SQLite, Oracle SQL, Shell Scripting, HTML, CSS, JavaScript, SysML, Julia, Wolfram, ROS, Arduino C
- Softwares: Windows, Linux Ubuntu, MS Office (Word, Powerpoint, Excel, Project, Publisher, Visio), Libre Office, ANSYS WorkBench, ABAQUS, CAD(Catia V5), Inkscape, Flight Dynamics Modelling (Athena AVL, Tornado VLM, US Datcom, XFLR5), SIMULINK, Presentation (Beamer, Lucid Press, Lucid Charts), Astrometrica, UNITY Game Development, VIM, Capella/ARCADIA (System Engineering), Valispace, Pschopy, Siemens TIA Portal
- Operating Systems: Unix/Linux, Windows, MacOS.

**ACADEMIC
EXPERIENCE**

Georgia Institute of Technology, Atlanta, Georgia US
Graduate Student Researcher

Oct, 2024 - Present

- [1] Exploring theoretical boundaries of Nonlinear Global Model Reduction as part of MAST Lab. (Jan, 2026 - Present)
- [2] Researching formulation of optimal control theory for a class of impulsively driven systems, using Distribution theory as part of MAST Lab. (Sep, 2025 - Dec, 2025)
- [3] Working and researching the implementation of optimal control theory on the class of Positive and Compartmental Systems as part of CASCADES Lab. (Feb, 2025 - Aug, 2025)
- [4] Worked as part of the Human Robot Collaboration Project at ROBOTICS Lab. Conducting Real-time experiments on the Manipulator arm collaboration with a Human and another robotic agent (Summer 2025)

Institute of Space Technology, Islamabad, Islamabad Capital Territory PK
Teaching Assistant

Feb, 2022 - June 2022

Assisting in Preparation and Deliverance of Lectures along with Making and Grading of Assessments.

PUBLICATIONS

- [1] M. N. Rauf and E. Verriest, "Generalized Euler-Lagrange Jump Conditions for Hybrid Systems," SoutheastCon 2026, Huntsville, AL, USA, 2026, pp. 1-5, doi: 10.1109/Southeast-Con63549.2026.11476625.
- [2] M. N. Rauf, E. Verriest, "Causal Infinitesimal Modeling and the Generalized Euler-Lagrange Equation in Impulsive Optimal Control", (*Under Review*).
- [3] M. N. Rauf, R. A. Khan, S. I. A. Shah, and M. A. Naqvi, "Design and analysis of stability and control for a small unmanned aerial vehicle," International Journal of Dynamics and Control, Nov 2023.
- [4] M. N. Rauf, R. A. Khan, and S. I. Ali Shah, "Stability analysis & development of a control strategy for a small uav," in 2023 3rd International Conference on Digital Futures and Transformative Technologies (ICoDT2), pp. 1-6, 2023.
- [5] M. N. Rauf, R. Khan, and G. Majeed, "Designing an Analysis Framework for estimating life of a component subjected to a constant load and a single overload using MATLAB," Journal of Material Science & Applied Engineering, pp. 1-10, 2022.

CONFERENCES
ATTENDED

- [1] “Stability Analysis & Development of Control Strategy for Small UAV”, *International Conference on Digital Futures and Transformative Technologies*, Islamabad, Islamabad Capital Territory, PK

PROFESSIONAL
EXPERIENCE

4Earth Inc., Kennesaw, GA, United States

Automation & Controls Engineer

Jul, 2025 - Aug, 2025

Duties: Responsible for assisting in CAD modeling, Design improvements of the Water Treatment unit.

Programming of Siemens Programmable Logic Controllers (PLC), and Siemens TIA Portal and Human-Machine Interfaces (HMI)

Projects:

- [1] Design and Implementation of an Autonomous Water Treatment unit *July, 2025 - Aug, 2025*
Suggested improvements in the design and systems integration leading to 10% improvement in Accessibility and Data Transparency.

Naqcode Technologies Pvt. Ltd., Islamabad, Islamabad Capital Territory PK

Team Lead (Flight Mechanics and Propulsion)

Sep, 2023 - Jul, 2024

Duties: Managed the Technical Project Execution of both Propulsion and Controls teams; Projects included Design and Development of SRMs, Modeling, Control & Simulation of UAVs, Designing of a UAV launch Platform.

Technical Execution and oversight of Flight Mechanics Projects, Modeling Framework design and Simulations.

Projects:

- [1] Design of Vehicle-Top UAV Launcher as well as a Catapult Launch mechanism *May, 2024 - July, 2024*
Initial Schematic Design, Structural Analysis, and Integration.
- [2] Development of a Warhead system *Nov, 2023 - Mar, 2024*
Warhead System Design; Detonation Point Monitoring.
- [3] Design, Performance analysis, and Optimization of SRM *Nov, 2023 - Jul, 2024*
Design and Analysis of Missile Propulsion system; Parametric studies of SRM metrics; Optimization of design variables via Genetic Algorithms.
- [4] FDM and Control of Precision Glide Vehicle *Mar, 2023 - Jan, 2024*
Development of a suitable flight control law for a Precision Guided System; Structural Sizing of Control Surfaces.
- [5] Design and Development of deployable SUCAV *Aug, 2022 - Jul, 2024*
Assessment of Stability; Development of Flight Dynamics Model

Flight Mechanics & Control Engineer

Aug, 2022 - Aug, 2023

Duties: Flight Dynamics modeling, Stability and Controllability assessment for Unconventional Vehicles; Mission Designing, and developing Guidance, & Navigation strategies; Control Systems Architecture design including Stability Augmenters, and Autopilots; Ensuring compliance of built systems with Established Standards.

Projects:

- [1] FDM and Control Development of Loitering Munition *Aug, 2022 - Nov, 2023*
Assessment of Stability; Development of Flight Dynamics Model; Designing and Verifying the Control Systems Architecture for 6DOF Control; Software Designing of Navigation Autopilots.

OTHER
EXPERIENCES

GT Boeing

Mentee

May, 2025 - Sep, 2025

Duties: Mentee under [Hunter Kim](#) in GT Boeing FLT Program 2025

Achieved Academy SGAC

EPD Team Member

Feb, 2025 - Present

Duties: Working as part of the Communication team in Education and Professional Development Sector of SGAC lead by Achieved Academy

All Pakistan Asteroid Search Campaign, Islamabad, Islamabad Capital Territory PK

Co-Founder

Aug, 2021 - Present

Duties: Self Launched campaign in affiliation with International Astronomical Search Collaboration and NASA; Technical Writing Lead.

Projects:

- [1] APASC Campaign 2024 *Sep, 2023 - Oct, 2023*
Increased Teams to 15 with 60 participants in total.
- [2] APASC Campaign 2023 *Sep, 2023 - Oct, 2023*
Collaborated with SGAC and Attock Astronomical Society to Launch the campaign nation wide.
- [3] APASC Campaign 2022 *Sep, 2022 - Oct, 2022*
Our Pilot campaign launched through Space Society IST in collaboration with Attock Astronomical Society.

GT CRAB Lab Robophysics Bootcamp

Participant

July, 2025 - July, 2025

Duties: Selected as one of the 10 participants in the 2025 rendition of the Robophysics Bootcamp organized by **CRAB Lab**. Learned about principles of physics-informed robot design and designed a 4-link snake-emulating robot.

Space Generation Advisory Council

Mentor

Dec, 2024 - May, 2025

Duties: Participated in the 2024 Mentorship Program, and served as an Academic and Professional Mentor to **Carl Anthony Sta Ana**

Aviation Training Hub, Islamabad, Islamabad Capital Territory PK

Aviation Intern

Nov, 2021 - Jan, 2022

Duties: Aviation Maintenance Training

Kamra Aeronautical Complex, Kamra PK

Aerospace Intern

Aug, 2021 - Sep, 2021

Duties: Observing and Learning the design, and production of trainer and other aircrafts.

REFERENCES

Available on request

PROFESSIONAL AFFILIATIONS

- I. Techmasters (Toastmasters International) Club *Apr, 2025 - Present*
- II. Honor Society *Feb, 2025 - Present*
- III. The Planetary Society *Jan, 2025 - Present*
- IV. Order of the Engineers *Dec, 2024 - Present*
- V. The American Society of Mechanical Engineers (ASME) *Sep, 2024 - Present*
- VI. Institute of Electrical and Electronics Engineers (IEEE) *Sep, 2024 - Present*
- VII. Royal Aeronautical Society *Aug, 2024 - Present*
- VIII. Fulbright Association *May, 2024 - Present*

IX. American Society for Testing and Materials	May, 2024 - <i>Present</i>
X. Institution of Engineers Pakistan	May, 2024 - <i>Present</i>
XI. European Low Gravity Research Association	Feb, 2024 - <i>Present</i>
XII. Moon Village Association	Apr, 2023 - <i>Present</i>
XIII. Space Generation Advisory Council	Feb, 2023 - <i>Present</i>

EXTRACURRICULAR **GT Buzz Buddies**, Atlanta, GA, USA

Mentor July, 2025 - Dec, 2025

Moon Village Association, International

Project Member of PESC project Apr, 2023 - *Present*

Space Generation Advisory Council, International

Mentor at 2024 SGAC Mentoring Program Dec, 2024 - May, 2025

Member of Space Exploration Project Group Feb, 2024 - *Present*

Project Member in Small Satellite Group Feb, 2023 - *Present*

Space Society IST (Publications Department), Institute of Space Technology, PK

Director Jan, 2022 - Sep, 2022

Google Development Student Club (IST-Chapter), Institute of Space Technology, PK

Executive Member (Publication) Sep, 2021 - Sep, 2022

Space Summer School, Institute of Space Technology, PK

Team Member (Student Management Team) July. 2022

Seventh ICASE Conference, Institute of Space Technology, PK

Team Manager (Publication) Dec. 2021